



Installation Conditions

Gas engine-generator sets
with **Series 400**, **Series 500**, and **Series 4000** engines

MS65032/07E

Product name	Engine model	Application *		Product name	Engine model	Application *
mtu 6R400 GS	E3066Z5	3A		mtu 8V4000 GS	8V4000L32	3A
mtu 6R400 GS	B3066Z8	3A		mtu 8V4000 GS	8V4000L33 **	3A
mtu 6R400 GS	E3066Lx	3A		mtu 8V4000 GS	8V4000L33	3A
mtu 6R400 GS	E3066LHx	3A		mtu 8V4000 GS	8V4000L64	3A
mtu 6R400 GS	E3066D3	3A		mtu 8V4000 GS	8V4000L64FNER	3A
mtu 6R400 GS	E3066DH3	3A		mtu 12V4000 GS	12V4000L32	3A
mtu 6R400 GS	E3066D4	3A		mtu 12V4000 GS	12V4000L33 **	3A
mtu 6R400 GS	E3066DH4	3A		mtu 12V4000 GS	12V4000L33	3A
mtu 12V400 GS	E3042Z6	3A		mtu 12V4000 GS	12V4000L64	3A
mtu 12V400 GS	B3042Z7	3A		mtu 12V4000 GS	12V4000L64FNER	3A
mtu 12V400 GS	E3042Lx	3A		mtu 16V4000 GS	16V4000L32	3A
mtu 12V400 GS	E3042LHx	3A		mtu 16V4000 GS	16V4000L33 **	3A
mtu 12V400 GS	E3042D3	3A		mtu 16V4000 GS	16V4000L33	3A
mtu 12V400 GS	E3042DH3	3A		mtu 16V4000 GS	16V4000L64	3A
mtu 12V400 GS	E3042D4	3A		mtu 16V4000 GS	16V4000L64FNER	3A
mtu 12V400 GS	E3042DH4	3A		mtu 20V4000 GS	20V4000L32	3A
mtu 6R500 GS	E406-ct80	3A		mtu 20V4000 GS	20V4000L33 **	3A
mtu 6R500 GS	B406-ct135	3A		mtu 20V4000 GS	20V4000L33	3A
mtu 8V500 GS	E408-ct80	3A		mtu 20V4000 GS	20V4000L64	3A
mtu 8V500 GS	B408-ct135	3A		mtu 20V4000 GS	20V4000L64FNER	3A
mtu 12V500 GS	E412-ct80	3A				
mtu 12V500 GS	B412-ct135	3A				

* Continuous operation, unrestricted

** Reduced power

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to §15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

1 Preface		9 Lube Oil System, Lube Oil Circuit	
1.1 Preface	4	9.1 Extended lube oil system	33
2 Product Information		10 Exhaust System	
2.1 Product safety	5	10.1 Exhaust system	35
3 Requirements for Building and Foundation		10.2 Exhaust line	38
3.1 General requirements	6	10.3 Exhaust silencer	40
4 Room Information		10.4 Catalytic converter	41
4.1 Premises - Requirements	7	10.5 Exhaust gas heat exchanger	43
4.2 Installation compartment - Requirements	8	11 Insulation	
4.3 Switchgear room - Requirements	11	11.1 Insulation	44
4.4 Installation recommendations	12	12 Control	
5 Ventilation		12.1 Safety functions and protective circuits	45
5.1 Ventilation system	13	12.2 Quality of safety components	47
5.2 Combustion air supply	16	13 Generator Circuit Breaker	
5.3 Room-air supply	17	13.1 Selection criteria for generator circuit breaker	49
5.4 Room air flow rate	18	14 Mains Circuit Breaker	
5.5 Safety requirements for the ventilation system	19	14.1 Selection criteria for mains circuit breaker	50
6 Gas System		15 Cabling	
6.1 Customer-provided gas system	21	15.1 Electromagnetic compatibility and lines	51
6.2 Requirements for the customer-provided gas system	22	15.2 Cable routing	52
6.3 Required gas quality	23	15.3 Cable recommendations for the various line groups	53
7 Coolant Circuit, General		15.4 Cross-sectional areas of cables	56
7.1 Coolant circuits	24	15.5 Generator wiring	57
7.2 Application as pure electricity generator	25	15.6 Protective equipotential bonding and protective conductor	58
7.3 Application with heat recovery	26	15.7 Electrical integration (only applicable to Series 4000)	59
7.4 Application with exhaust gas heat recovery	27	16 Appendix A	
7.5 Coolant circuits - Design	28	16.1 Index	61
7.6 Emergency recoler	30		
8 Connection Regulations			
8.1 Pipes and connections	31		

1 Preface

1.1 Preface

These Installation Conditions form the basic principles for application engineers, system/component designers and assembly personnel for planning or carrying out the installation of mtu components.

The Installation Conditions are applicable to all systems and gas engines of Series 400, Series 500, and Series 4000 supplied by Rolls-Royce Solutions, see (→ Business Portal).

The Installation Conditions do not relieve those responsible for the system from independently ensuring that the equipment is installed and checked in an appropriate manner.

The following accompanying documents must be complied with, in addition to the Installation Conditions:

- Planning drawing
- Schematic diagrams
- TEN data
- Arrangement drawings
- Operating Instructions, see (→ Business Portal)
- Maintenance Schedule, see (→ Business Portal)
- Fluids and Lubricants Specifications, see (→ Business Portal)
- mtu standards

Refer to the specifications for the order concerned for information as to which documents apply.

Performing prescribed maintenance tasks properly and on schedule positively influences operational safety, reliability and service life. Ensure good accessibility for operation, maintenance and repair work when planning and installing the system.

Highlighted information

Important

This field contains product information which is important or useful for the user.

This information must not refer to hazards related to personal injury or material damage.

2 Product Information

2.1 Product safety

CE conformity of products from Rolls-Royce Solutions

The products from Rolls-Royce Solutions are designed and built in accordance with the Machinery Directive as well as other relevant EC directives. Rolls-Royce Solutions issues a “CE” declaration of conformity for complete machines according to the scope of delivery and provides such machines with a CE mark. For partly completed machines, a Declaration of Incorporation is issued.

In order to determine all hazards associated with a product, Rolls-Royce Solutions carries out a risk assessment. By implementing suitable protective measures, the hazards posed to human health and the environment are reduced or completely eliminated for all life stages in accordance with the state of the art. All important information on residual risks and on the safe and intended use of the machine can be found in the operating manual and other customer documents.

Integration of machinery / pressure devices / gas appliances

Rolls-Royce Solutions can issue declarations of conformity and incorporation only for scope of delivery. In order to determine possible safety-related linking between several machines / pressure devices / gas appliances, a higher-level risk assessment of the whole plant system is to be drawn up. The manufacturer of the complete plant is responsible for the preparation of the higher-order risk assessment, the implementation of the higher-order safety concept derived from it and the Declaration of Conformity with CE mark.

Depending on the plant system and the setup of the machine, some aspects may need to be considered:

- Internal and external explosions
- Fire
- Ventilation concept
- Heat utilization
- Biogas application
- Mechanical and electrical hazards
- Switching of the plant system, or a part thereof, to a safe state and protection against inadvertent start-up

The fulfillment of the conditions specified in this document is a prerequisite for the safe and smooth operation of the plant system.

Prior to the start of planning, an investigation into the requirements regarding official specifications has to be carried out and any required approvals obtained. The requirements for the building also apply to the container solutions, where appropriate.

Documents to be used:

- Technical description
- Technical data sheet
- Foundation data sheet
- P&I diagram
- Planning drawing
- Fluids and Lubricants Specifications
- Quality of safety components (→ Page 47)

3 Requirements for Building and Foundation

3.1 General requirements

The statics of the building must have a rating sufficient for the plant. If necessary, consult a structural engineer. Structural fire protection must be accomplished in accordance with the local laws and regulations. The foundation must be flat and such that it can safely support the weight of the engine-generator set and system components. The weight of the engine-generator set can be obtained from the technical data sheet. With regard to the vibrations caused by the engine-generator set, the foundation must be designed in accordance with the foundation data sheet.

Sufficient openings must be provided to install and remove components. The requirements of the louver dampers are described in the Ventilation chapter (→ Page 13). The building / container must have sufficient protection against lightning strokes. The plant must be integrated in the on-site lightning protection system. Specifications of insurance providers and authorities regarding lightning protection and fire protection must be researched (e.g. preparation of an integrated fire-protection concept, dividing into fire sections etc.).

Important

The technical data sheet and the foundation data sheet must be observed.

Important

To be observed in protection and danger zones (flooding areas):

When installing plants in protection and danger zones (flooding areas), general locally applicable laws, directives and standards must be observed.

4 Room Information

4.1 Premises – Requirements

To increase the level of work safety with regard to noise emission and vibration emission, it is recommended to install the engine-generator set and the switchgear in separate rooms, if possible. In so doing, the danger of the control cabinets overheating will be reduced. If a dedicated switchgear room cannot be provided, the installation site must comply with the requirements for the control cabinets delivered by Rolls-Royce Solutions.

Due to the noise emission, a sound-insulating enclosure is recommended for the engine-generator set. There must be sufficient lighting in the installation room and the room for the switchgear. The use of halogen spotlights with hot surfaces and high rates of heat loss is not permitted due to the risk of fire.

Emergency exits and escape routes must be well-lit and appropriately marked. Emergency exits and escape routes must not be blocked by cables or suchlike (pipes, hoses). The escape routes from other rooms must not pass through rooms for switchgear or the installation room. Escape doors must open in the direction of escape.

It must be possible to lock the installation room and the switchgear room, so that access by unauthorized persons can be prevented.

If the engine-generator set or the module is delivered by Rolls-Royce Solutions as a container solution, the container must be secured such that willful damage or unauthorized access by third parties can not occur.

4.2 Installation compartment - Requirements

For safe operation of the engine-generator set, the installation compartment must be suitably arranged. During the planning phase, you should already check whether compliance with further official requirements is necessary.

The installation compartment must be large enough to permit easy operation and maintenance of the module / engine-generator set as well as sufficient ventilation. The dimensions of the engine-generator set are shown in the planning drawing. The space required for movement during operation, maintenance and repair is also shown in the planning drawings. These dimensions required for movement must be observed in all circumstances. In addition to the dimensions of the engine-generator set and space required for movement, the space requirements of the peripheral connected systems (coolant circuits, exhaust system, oil system) must also be taken into consideration. Surfaces subject to heating must be insulated. The space required for insulation, the required minimum distance to walls and between the individual objects must be guaranteed. The erection of climbing aids must be possible. Ventilation ducts must be at least 0.7 m above the top edge of the mixture cooler.

Important

Comply with the planning drawing.

Lifting tools for maintenance work

Lifting equipment suitable for the heavy component parts of the engine and the exhaust system (turbocharger, mixture cooler, SCR element, etc.) must be provided in the installation compartment. For the maximum load, refer to the arrangement drawing of the engine-generator set.

Adhere to the applicable regulations and standards of the country of installation.

The installation of a crane rail or a mobile crane is necessary for the assembly work. A stop must be provided to limit the movement on removal beams and crane rails.

Important

Observe the planning drawing and the technical data sheet.

Fire protection requirements

Potential fire loads in the installation compartment must be minimized. In particular, no combustible material must be deposited or stored in the installation compartment. The minimum distance required for fire protection reasons between pipelines, other components and walls subject to heating must be observed. Wall ducts with pipelines and cables must be sealed off accordingly. Measures must be introduced for early detection of fires and smoldering fires, e.g. the installation of a fire detection system. Suitable extinguishing agents must be available (fire extinguisher, fire extinguishing system). No combustible material must be deposited or stored in the installation compartment.

The corresponding national legislation and local regulations on fire protection must be observed.

Explosion protection (OEG)

The Fluids and Lubricants Specifications must be observed.

The gas supply system before the inlet to the gas train must be designed to prevent the ingress of oxygen. If necessary, process-related access points such as condensate traps must be monitored.

According to the state of technology, suitable measures must be introduced because the possibility of an explosion in the exhaust system can not be excluded.

More detailed requirements are described in the Exhaust system chapter.

Exterior explosion protection (UEG)

The gas supply, including gas train, in the installation compartment must have a leaktight design and must be checked on a regular basis for continuous leak-tightness. With circulation air systems, an exhaust air duct must be provided through which gas, in the event of a gas leakage, can be safely discharged. As a supplementary protective measure, the installation of a gas alarm system to be integrated in the plant control system is necessary. For this purpose, at least one gas warning sensor must be installed above the gas engine in direction exhaust air. Dead corners in the installation compartment without a flow should be avoided where possible because gas from leaks can accumulate there. If this is no longer possible for design-related reasons, gas warning sensors should be installed here. The room ventilation must be dimensioned such that, in the event of leakages on gas-carrying parts, a potentially explosive atmosphere can not be created. The gas-air mixture must be safely discharged directly to the atmosphere. For more detailed requirements of the ventilation system, refer to the Ventilation chapter.

Noise protection

Exact details on noise emission can be obtained from the technical data sheet. In accordance with these values, suitable measures must be taken, e.g. noise insulating paneling of the engine-generator set (acoustic enclosure) or acoustic boards in the installation compartment. The respective national legislation and local regulations must be observed.

Vibrations

The engine-generator set must be mounted on insulating boards to reduce the vibration transfer to the building and foundation. The length and number of the boards to be provided depends on the size and weight of the engine-generator set according to the planning drawing from Rolls-Royce Solutions. External vibrations must not act on the engine-generator set because this affects the natural frequency of the engine-generator set, which can be increased. This can result in failure or breakage of components.

Frost protection

The temperature in the installation compartment must not fall below 5 °C in any section. It may be necessary to install a heater.

Environmental protection

The installation compartment must comply with local water protection regulations (e.g. VAWS - Ordinance on Installations for the Handling of Substances Hazardous to Water and on Specialist Firms) and the WHG (Water Resources Act). If an extended oil system is used, the tank(s) must have a double-walled design and be equipped with a leakage indicator system. To reliably prevent the penetration of fluids and lubricants and water-polluting substances in the water system / groundwater, local regulations and requirements (oil separator, retaining device, leakage detection) must be observed.

Any other possible specifications (installation of an oil separator or a neutralization system for accumulated exhaust gas condensate) must be requested from the local authorities.

When installing or erecting plants in protection and danger zones, additional country-specific laws, directives and standards must be observed.

Ambient conditions (dust, moisture)

If control cabinets are located in the installation compartment, the ambient conditions must comply with the design of the switchgear/control cabinets, (→ Page 11).

Equipotential bonding, protective conductor and grounding

For the selection and installation of electrical equipment, grounding arrangements and protective conductors, see DIN VDE 0100-540. Since the design of the equipotential bonding system and the protective conductor depends on the local mains type, details can not be discussed here. Only generally applicable information independent of the form of power grid can be provided at this point.

Equipotential bonding

Apart from the module frame and heat module frame, all other metallic, electrically conductive components must be included in the local equipotential bonding. This includes heating water pipes, gas line, exhaust system and cable trays. Several connection points for equipotential bonding have already been prepared at the module frame. The equipotential bonding is implemented from the module frame to the generator frame via flexible connecting lines.

Protective conductor

The protective conductor must be connected in every case to the housing of the generator. Grounding of the generator neutral point depends on plant-specific factors. The local regulations of the power supply company or the safety regulations must be observed to ensure that grounding is carried out correctly.

ZEP - Central grounding point

State of the art is the version with a central grounding point. This reliably prevents stray currents. If a central grounding point is used, there is only one connection between PEN (protective conductor with neutral conductor function) and ground. The complete run of the PEN conductor must be insulated. For networks with decentralized power supply, a further central ground point is possible. The two areas with central grounding points must not be operated in parallel. A central grounding point must be provided in the installation compartment.

Installation compartment warnings

The installation contains electromagnetic fields. For this reason, persons with implanted medical devices, e.g. pacemakers, must not enter the installation compartment and switchgear rooms. Appropriate warnings must be affixed at the doors to the switchgear room and installation compartment. The delivery from Rolls-Royce Solutions includes a set of "Warnings" stickers for the installation compartment.

Further instruction plates

An instruction plate for safe conduct in the event of a fire or gas alarm must be provided. Also required is an instruction plate noting that the ventilation powers up in the event of simultaneous gas alarm and manual emergency stop.